

Probability

Why Study Probability?

Probability theory is an important branch of mathematics useful in daily life decision making as well as all scientific and engineering endeavors. It is a fundamental way of thinking and problem solving skill.

“We see that the theory of probability is at bottom only common sense reduced to calculation; it makes us appreciate with exactitude what reasonable minds feel by a sort of instinct, often without being able to account for it. ... It is remarkable that this science, which originated in the consideration of games of chance, should have become the most important object of human knowledge. ... The most important questions of life are, for the most part, really only problems of probability.” Simon de Laplace (1749-1827)

Definition of Probability

“To study a phenomenon, one must reduce all events of the same type to a certain number of **equally possible cases**, and then the **probability** is a fraction whose numerator represents the number of cases favorable to the event and whose denominator represents the number of possible cases.” Simon de Laplace (1749-1827), *Théorie Analytique des Probabilités*

$$P(event) = \frac{\text{number of favorable cases}}{\text{number of possible cases}}$$

Interpretation of Probability

- **Frequency interpretation:** through repeated experiments, the probability of the outcome is the proportion of the experiments that result in the outcome. Usually used by scientists
e.g. $P(\text{a fair die will roll a 6}) = \frac{1}{6}$

You can repeat experiment a lot of times to get the number 1/6

- **Subjective interpretation:** a belief of the chance that an outcome will happen. Usually used by philosophers or decision maker
e.g. $P(\text{person x will become the chief executive}) = \frac{1}{3}$

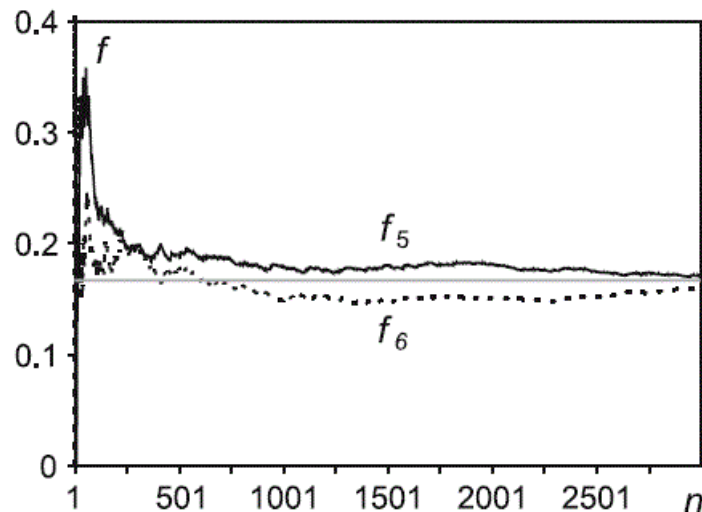
The event either occurs or not occur. You cannot repeat the experiment

Example

How would you interpret the following using the frequency interpretation and the subjective interpretation?

$$P(\text{tomorrow will be a rainy day}) > \frac{1}{5}$$

$$P(\text{a 5 and then a 6 is obtained in two dice rolls}) = \frac{1}{36}$$



Frequency curves of throwing '5' and '6' in a real dice

Example: (It is important to divide into equally possible cases)

Consider a family with two children. Assume

$$P(\text{boy}) = P(\text{girl}) = 0.5$$

What is the probability of the family with one boy and one girl?

One may intuitively divide into three cases:

Case A: the family has two boys

Case B: the family has one boy and one girl

Case C: the family has two girls

Then $P(\text{one boy and one girl}) = \frac{1}{3}$ which is wrong! Why?

One should divide into equally possible cases, or elementary events.
These are events with equal probability to occur.

$\{MM, MF, FM, FF\}$

M: Male F: Female

Hence $P(\text{one boy and one girl}) = \frac{1}{2}$

(Name a hidden assumption behind this model)

In general, we need to have a language to reason about events and their interaction with each other. Below is such a language

Sample Space and Events

- **Sample Space** S : The set of all possible outcomes of an experiment

E.g. The sample space of a roll of fair dice is

$$S = \{1, 2, 3, 4, 5, 6\}$$



- **Event** E : Any subset of the sample space. An event is a set consisting of possible outcomes of an experiment.

E is the event that a dice roll gives an even number

$$E = \{2, 4, 6\}$$

- **Union** of two events $E \cup F$

F is the event of all dice roll gives a number larger than 3

$$F = \{4, 5, 6\}$$

$E \cup F$ is the event that “the dice roll gives an even number or a number larger than 3”

$$E \cup F = \{2, 4, 5, 6\}$$

- **Intersection** of two events $E \cap F$

$E \cap F$ is the event that “the dice roll gives an even number larger than 3”

$$E \cap F = \{4, 6\}$$

Sometimes, we write $E \cap F$ as EF

- **Complement of an event** : The complement of an event E , denoted E^c , consists of all outcomes in the sample space S that are not in E

$$E^c = \{1,3,5\}$$

As another example, if $E = \{1\}$ (the event that the dice rolls '1'), then $E^c = \{2,3,4,5,6\}$ (the event that the dice does not roll '1')

- E is **contained** in F if any outcome in E is also an outcome in F . We denote this as $E \subseteq F$ (In the book, $\subset$ is used)

If $E = \{2,4,6\}$ and $F = \{2,3,4,5,6\}$, $E \subseteq F$

- E and F are **equal** if $E \subseteq F$ and $F \subseteq E$

- **Mutually exclusive events** : Two events E and F are mutually exclusive if $E \cap F = \emptyset$ ($\emptyset$ denotes empty set)

E : the dice roll gives an even number $E = \{2,4,6\}$

F : the dice roll gives an odd number $F = \{1,3,5\}$

$$E \cap F = \emptyset$$

- **We may denote more than one events using similar notations, e.g.**

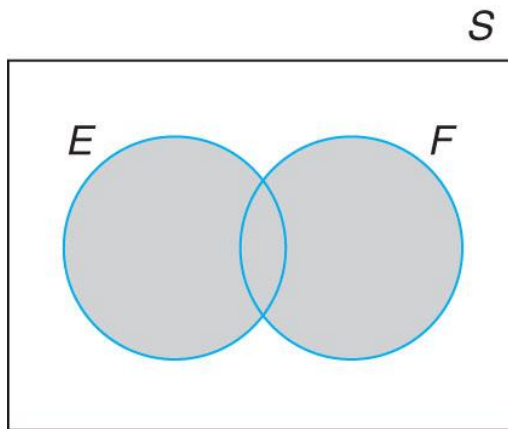
$$\bigcup_1^n E_i = E_1 \cup E_2 \cup \cdots \cup E_n$$

$$\bigcap_1^n E_i = E_1 \cap E_2 \cap \cdots \cap E_n$$

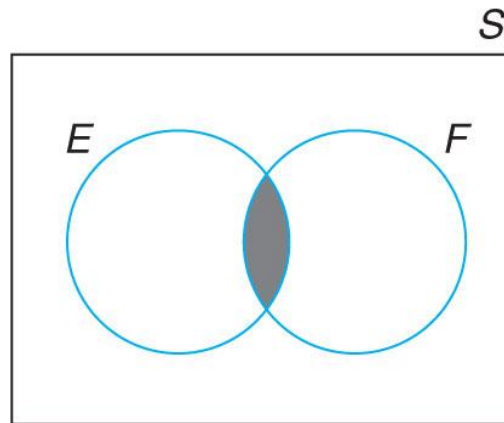
Venn Diagram

Invented by the 19th century English mathematician John Venn

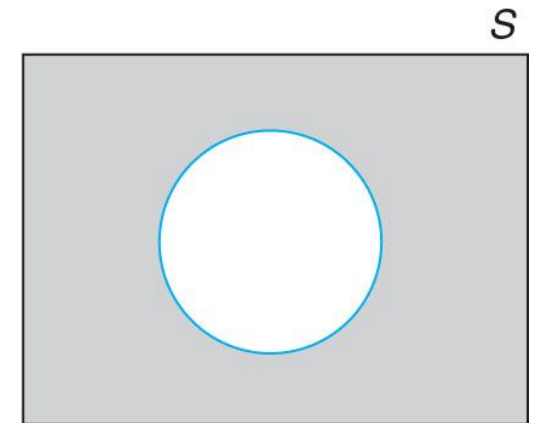
Represent events and their interactions by circles. We can use it to prove various theorems about events.



(a) Shaded region: $E \cup F$



(b) Shaded region: EF



(c) Shaded region: E^c

Theorems about Events

These theorems are common to both probability theory and logic theory

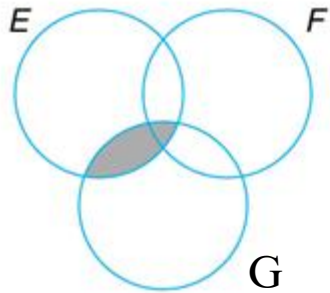
Commutative Law $E \cup F = F \cup E$ $EF = FE$

Associative Law $(E \cup F) \cup G = E \cup (F \cup G)$
 $(EF)G = E(FG)$

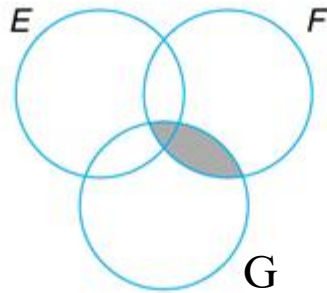
Distributive Law $(E \cup F)G = EG \cup FG$
 $EF \cup G = (E \cup G)(F \cup G)$

DeMorgan's Laws $(E \cup F)^c = E^c F^c$
 $(EF)^c = E^c \cup F^c$

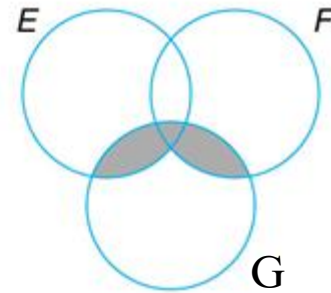
Example: Proving distributive law $(E \cup F)G = EG \cup FG$



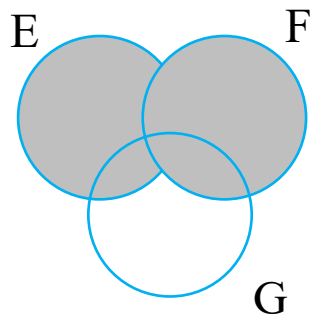
EG



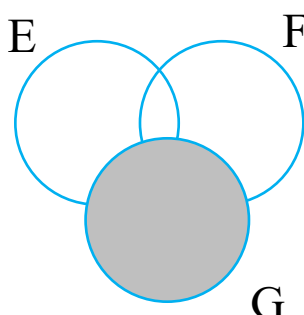
FG



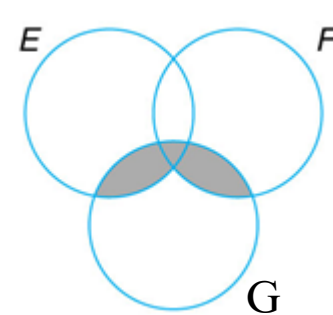
$EG \cup FG$



$E \cup F$



G



$(E \cup F)G$

Another way to prove the law is by enumerating all cases using truth tables

E	F	G	EG	FG	$E \cup F$	$EG \cup FG$	$(E \cup F)G$
T	T	T	T	T	T	T	T
T	T	F	F	F	T	F	F
T	F	T	T	F	T	T	T
T	F	F	F	F	T	F	F
F	T	T	F	T	T	T	T
F	T	F	F	F	T	F	F
F	F	T	F	F	F	F	F
F	F	F	F	F	F	F	F

↑ ↑
same truth values

Axioms of Probability

- One can derive the probability theory from the axioms below:
- AXIOM 1 $0 \leq P(E) \leq 1$
- AXIOM 2 $P(S) = 1$
- AXIOM 3

For any sequence of mutually exclusive events $E_1, E_2, \dots$

$$P(\cup_{i=1}^n E_i) = \sum_{i=1}^n P(E_i)$$

Using Axioms to Prove Propositions

Proposition: $P(E^c) = 1 - P(E)$

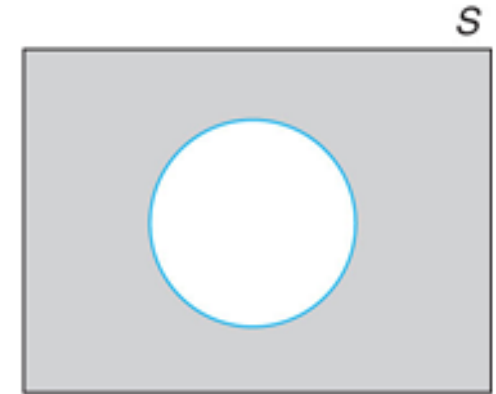
Proof:

By Axiom 2, $1 = P(S)$

Since $S = E \cup E^c$, $P(S) = P(E \cup E^c)$

By Axiom 3, $P(E \cup E^c) = P(E) + P(E^c)$

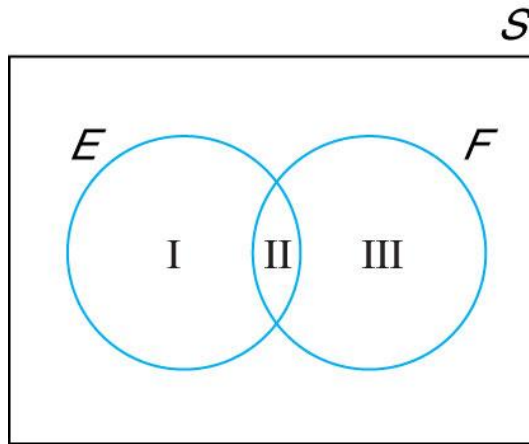
Hence $P(E^c) = 1 - P(E)$



shaded region is
 E^c

Proposition: $P(E \cup F) = P(E) + P(F) - P(EF)$
(inclusion-exclusion principle)

Proof:



Intuition: because region II
is counted twice

As regions I, II, III are mutually exclusive. By Axiom 3

$$P(E \cup F) = P(I) + P(II) + P(III)$$

$$P(E) = P(I) + P(II)$$

$$P(F) = P(II) + P(III)$$

Thus

$$P(E \cup F) = P(E) + P(F) - P(II) = P(E) + P(F) - P(EF)$$

Example

In a roll of fair dice, let E be the event the number is even. Let F be the event that the number is less than or equal to 4. What is the probability that the number is even or less than or equal to 4?

$$E = \{2, 4, 6\}$$

$$F = \{1, 2, 3, 4\}$$

$$EF = \{2, 4\}$$

$$P(E) + P(F) = \frac{1}{2} + \frac{4}{6} > 1 \quad \text{Wrong!}$$

$$P(E \cup F) = P(E) + P(F) - P(EF) = \frac{1}{2} + \frac{4}{6} - \frac{2}{6} = \frac{5}{6}$$

Example

Chevalier de Méré, a writer and nobleman from Louis XIV's court, found that it is a good strategy to bet that a '6' would turn up with one die in 4 throws, but he constantly lost if he betted that two '6' would turn up with two dice in 24 throws. He asked Pascal why? Can you help him?

Hint: You have to use the complement cleverly

$$P(\text{no } 6) = 1 - \frac{1}{6} = \frac{5}{6}$$

Since the number obtained by a dice in one throw and that obtained in another throw should be “**independent**”

$$P(\text{no } 6 \text{ in } 4 \text{ throws}) = \left(\frac{5}{6}\right)^4$$

Using the complement proposition: $P(E^c) = 1 - P(E)$

$$\begin{aligned} P(\text{at least one 6 in 4 throws}) &= 1 - P(\text{no 6 in 4 throws}) \\ &= 1 - \left(\frac{5}{6}\right)^4 = 0.51775 > 0.5 \end{aligned}$$

Therefore he usually wins

Using two dice, **try using the same argument**,

$$\begin{aligned} P(\text{no double 6 with two dice}) &= 1 - \frac{1}{36} = \frac{35}{36} \\ P(\text{no double 6 in 24 throws}) &= \left(\frac{35}{36}\right)^{24} \end{aligned}$$

$$P(\text{at least one double 6 in 24 throws}) = 1 - \left(\frac{35}{36}\right)^{24} = 0.49140 < 0.5$$

Therefore he usually loses

Sample Spaces Having Equally Likely Outcomes

- For a large number of experiments, it is natural to assume that each point in the sample space is equally likely to occur:

$$S = \{1, 2, \dots, N\}$$

$$P(\{1\}) = P(\{2\}) = \dots = P(\{N\}) = p$$

By Axiom 2 $1 = P(S)$

By Axiom 3

$$P(S) = P(\{1, 2, \dots, N\}) = P(\{1\}) + \dots + P(\{N\}) = Np$$

Thus $P(\{i\}) = p = \frac{1}{N}$

By Axiom 3, for any event E

$$P(E) = \frac{\text{number of points in } E}{N}$$

- So we need **effective ways to count** the number of ways a given event can occur

Basic Principle of Counting

- The mathematical subject of counting is called “combinatorics”
- Suppose that two experiments are to be performed. Then if experiment 1 can result in any one of m possible outcomes and if, for each outcome of experiment 1, there are n possible outcomes of experiment 2, then together there are mn possible outcomes of the two experiments
- The basic principle can be proven by enumerating all the possible outcomes of the two experiments:

$(1,1), (1,2), \dots, (1,n)$

$(2,1), (2,2), \dots, (2,n)$

$\vdots$

$(m,1), (m,2), \dots, (m,n)$

There are m rows and n columns

Example

Two dice is rolled. The first dice is odd and the second dice is even. How many possible ways of throwing the dice are there?

The first dice may be $\{1, 3, 5\}$. The second dice may be $\{2, 4, 6\}$.
The possible ways are

$(1, 2), (1, 4), (1, 6)$

$(3, 2), (3, 4), (3, 6)$

$(5, 2), (5, 4), (5, 6)$

or 9 possibilities

Generalized Basic Principle of Counting

- If r experiments that are to be performed are such that the first one may result in any of n_1 possible outcomes, and if for each of these n_1 possible outcomes there are n_2 possible outcomes of the second experiment, and if for each of the possible outcomes of the first two experiments there are n_3 possible outcomes of the third experiment, and if, ..., then there are a total of $n_1 \cdot n_2 \cdots n_r$ possible outcomes of the r experiments

Example

A password has 8 letters. The first letter must be Capital Letter and the next 4 letters are lower case letters. The last three letters are numbers. How many different passwords are possible?

The first letter may be 'A' to 'Z'. There are 26 possibilities.

Similarly, the second letter may be 'a' to 'z', which has 26 possibilities.

The numbers may be '0' to '9'. There are 10 possibilities.

Thus the number of different passwords are

$$26 \times 26^4 \times 10^3 \approx 1.188 \times 10^{10}$$

Permutation

- Number of different ways n distinct objects can be arranged in a linear order
- Let the n objects be $1, 2, \dots, n$
- In the first position, we have n choices. In the second position, since we have already made a choice in the first position, we have $(n - 1)$ choices, ...
- There are $n(n - 1) \cdots 2 \cdot 1$ ordered arrangement
- Each ordered arrangement is called a **permutation**
- Define $n! = n(n - 1) \cdots 2 \cdot 1$ $n!$ is called “ n factorial”
- $2! = 2 \cdot 1 = 2$ $1! = 1$
- Define $0! = 1$ (this definition will be useful for later calculations)

Example

Suppose you have five books. In how many ways can the books be put on the bookshelf?

For the first book to be put on the left, we have 5 choices. After you have selected the first book, for the second book, we have four choices, and so on.

Hence there are $(5)(4)(3)(2)(1) = 5!$ ways.

Combination

Suppose now that we are interested in determining the **number of different groups** of r objects that could be formed from a total of n objects. For instance, how many different groups of three could be selected from the five items A, B, C, D, E ? To answer this, reason as follows. Since there are 5 ways to select the initial item, 4 ways to select the next item, and 3 ways to then select the final item, there are thus $5 \cdot 4 \cdot 3$ ways of selecting the group of 3 when the order in which the items are selected is relevant. However, since every group of 3, say the group consisting of items A, B, C , will be counted 6 times (that is, all of the permutations $ABC, ACB, BAC, BCA, CAB, CBA$ will be counted when the order of selection is relevant), it follows that the total **number of different groups** that can be formed is $(5 \cdot 4 \cdot 3)/(3 \cdot 2 \cdot 1)$
 $= 10$

- In general, as $n(n - 1) \cdots (n - r + 1)$ represents the number of different ways that a group of r items could be selected from n items **when the order of selection is considered relevant**, and since **each group** of r items will be counted $r!$ Times in this count, the number of **different groups of r items** that could be formed from a set of n items is

$$\frac{n(n - 1) \cdots (n - r + 1)}{r!} = \frac{n!}{(n - r)! r!}$$

- Define the **number of combinations** of n objects taken r at a time as

$$\binom{n}{r} \quad (r \leq n)$$

$$\binom{n}{r} = \frac{n!}{(n - r)! r!}$$

- Define $\binom{n}{0} = \binom{n}{n} = 1$

Example

There are 120 students in Grade 6 of your secondary school. A Student Committee of 3 students is to be elected from Grade 6 students. How many different possibilities for the composition of the Committee is there?

There are 120 possibilities to choose the first student, 119 possibilities to choose the second student, 118 possibilities to choose the third student. Hence there are $(120)(119)(118)$ possibilities.

However, suppose (Jean, Jane, Janet) are the three students in a committee, then (Jean, Janet, Jane), (Jane, Jean, Janet), (Jane, Janet, Jean), (Janet, Jean, Jane), (Janet, Jane, Jean) are the same committee. Each of the $3!$ possibilities actually represent the same committee. Hence the number of different committees are

$$\frac{(120)(119)(118)}{3!} = \frac{120!}{3! 117!} = \binom{120}{3}$$

Example

Suppose 8 students are divided into 2 freely formed groups of 4. How many possible grouping arrangements are there?

The first group has 4 students. There are $\binom{8}{4}$ ways of forming the first group. After the first group is formed, the remaining students form the second group. Hence there are

$$\binom{8}{4} = \frac{8!}{4! 4!} = 70$$

arrangements. As the question does not say that group names are important, there are only 35 grouping arrangements

Example

‘Mark Six’ is a well-known betting game in Hong Kong run by the Hong Kong Jockey Club. Calculate the winning probabilities for ‘Mark Six’. [Mark Six rules: 49 numbers (‘1’ to ‘49’). Draw 6 numbers randomly without replacement. Then draw an extra number]

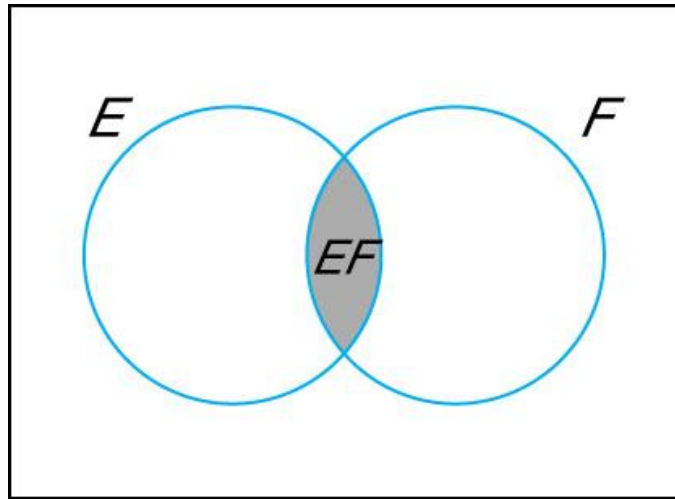
- 1st Prize All 6 drawn numbers are correct
- 2nd Prize 5 out of 6 drawn numbers correct + the extra number correct
- 3rd Prize 5 out of 6 drawn numbers correct
- 4th Prize 4 out of 6 drawn numbers + the extra number correct
- 5th Prize 4 out of 6 drawn numbers correct
- ⋮

Conditional Probability

We are often interested in calculating probabilities when some partial information concerning the result of the experiment is available, or in recalculating them in light of additional information

Suppose that one rolls a pair of a fair dice. The sample space S of this experiment is $S = \{(i, j), i = 1, 2, 3, 4, 5, 6, j = 1, 2, 3, 4, 5, 6\}$

Suppose F is the event that the first roll is 3 (condition). Let E be the event that the sum of the two dice rolls is 7. Then there is a 1 in 6 chance that the second roll gives a 4, so that the sum of the two dices is 7. Hence the conditional probability is $\frac{1}{6}$. We denote the conditional probability as $P(E|F)$ (probability of E given that F has occurred)



- If the event F occurs, then in order for E to occur it is necessary that the actual occurrence be a point in both E and F ; that is, it must be in EF . However, because we know that F has occurred, it follows that we can regard F as the new sample space and hence the probability that the event EF occurs will equal the probability of EF relative to the probability of F . That is

$$P(E|F) = \frac{P(EF)}{P(F)}$$

- $P(F) > 0$ for the above equation to be well defined

Example

A fair coin is one for which there is a equal probability of turning up “head” and “tail”. If a fair coin is tossed twice, what is the conditional probability that both flips land on “heads”, given that the first flip lands on “heads”?

Let $E = \{(h, h)\}$ be the event that both flips land on “heads”.

Let $F = \{(h, h), (h, t)\}$ be the event that the first flip lands “heads”.

$$P(E|F) = \frac{P(EF)}{P(F)} = \frac{P\{(h, h)\}}{P\{(h, h), (h, t)\}} = \frac{1}{2}$$

We have to identify the events correctly

Example

A student is taking a 2 hour exam. Suppose the probability that the student will finish the exam in less than x hours is $x/2.5$. Given that **the student is still working after 1.5 hours**, what is the conditional probability that **he can finish the exam within 2 hours**?

Let F be the event **the student is still working after 1.5 hours**. Then F^c is the event that the student finishes in less than 1.5 hours.

$$P(F) = 1 - P(F^c) = 1 - \frac{1.5}{2.5} = 0.4$$

Let E be the event that **he finishes the exam within 2 hours**. Then EF is the event that he is still working after 1.5 hours but finishes within 2 hours.

$$P(E|F) = \frac{P(EF)}{P(F)} = \frac{\frac{2}{2.5} - \frac{1.5}{2.5}}{0.4} = 0.5$$

Example

Joseph is undecided whether to take course A or course B. He estimates that his probability of getting B+ would be $\frac{2}{3}$ for course A and 0.4 for course B. If Joseph decides to base his decision on the flip of a fair coin, what is the probability that he gets a B+ in course A?

Let F be the event that Joseph takes course A and E be the event that he gets a B+.

$$P(EF) = P(E|F)P(F) = \left(\frac{2}{3}\right)(0.5) = \frac{1}{3}$$

Example

Suppose that a bag contains 4 red balls and 4 white balls. We draw 2 balls from the bag without replacement. If we assume that at each draw each ball is equally likely to be chosen, what is the probability that both balls drawn are white?

Let F be the event that the first ball is white and E be the event that the second ball is white. $P(F) = \frac{4}{8} = 0.5$.

Since the first ball is white, there are 4 red balls and 3 white balls left, hence $P(E|F) = 3/7$

$$P(EF) = P(E|F)P(F) = \left(\frac{3}{7}\right)(0.5) = \frac{3}{14}$$

For more than two events

$$\begin{aligned} &P(E_1 E_2 E_3 \cdots E_n) \\ &= P(E_1)P(E_2|E_1)P(E_3|E_1 E_2) \cdots P(E_n|E_1 \cdots E_{n-1}) \end{aligned}$$

Proof

Apply the definition of conditional probability to the right hand side

$$\begin{aligned} &P(E_1) \frac{P(E_1 E_2)}{P(E_1)} \frac{P(E_1 E_2 E_3)}{P(E_1 E_2)} \cdots \frac{P(E_1 E_2 \cdots E_n)}{P(E_1 E_2 \cdots E_{n-1})} \\ &= P(E_1 E_2 E_3 \cdots E_n) \end{aligned}$$

Example

- Estimating the probability of finding extraterrestrial intelligence. Apply the argument to the Drake equation

The Drake equation is:

$$N = R_{\star} \cdot f_p \cdot n_e \cdot f_l \cdot f_i \cdot f_c \cdot L$$

where:

N = the number of civilizations in our galaxy with which communication might be possible (i.e. which are on our current past light cone);

and

$R_{\star}$ = the average rate of star formation in our galaxy

f_p = the fraction of those stars that have planets

n_e = the average number of planets that can potentially support life per star that has planets

f_l = the fraction of planets that could support life that actually develop life at some point

f_i = the fraction of planets with life that actually go on to develop intelligent life (civilizations)

f_c = the fraction of civilizations that develop a technology that releases detectable signs of their existence into space

L = the length of time for which such civilizations release detectable signals into space^{[5][6]}